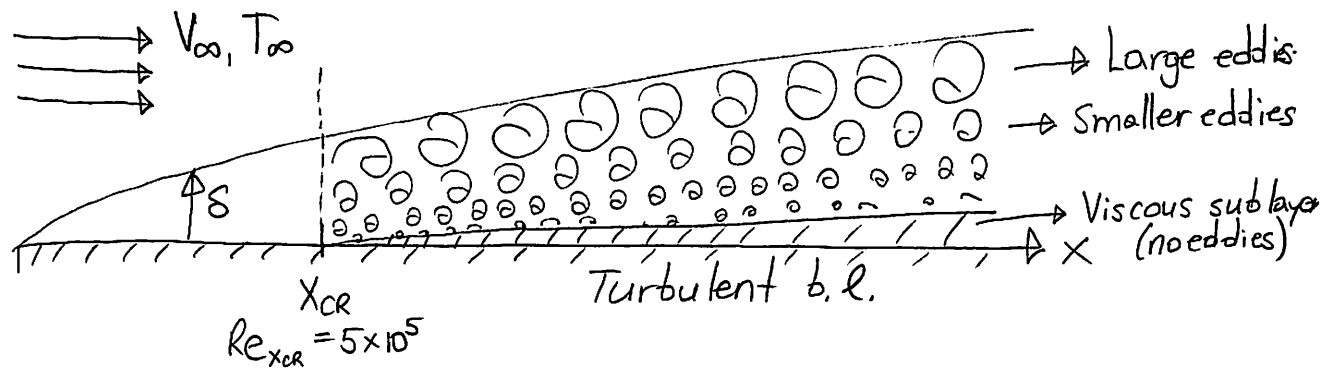


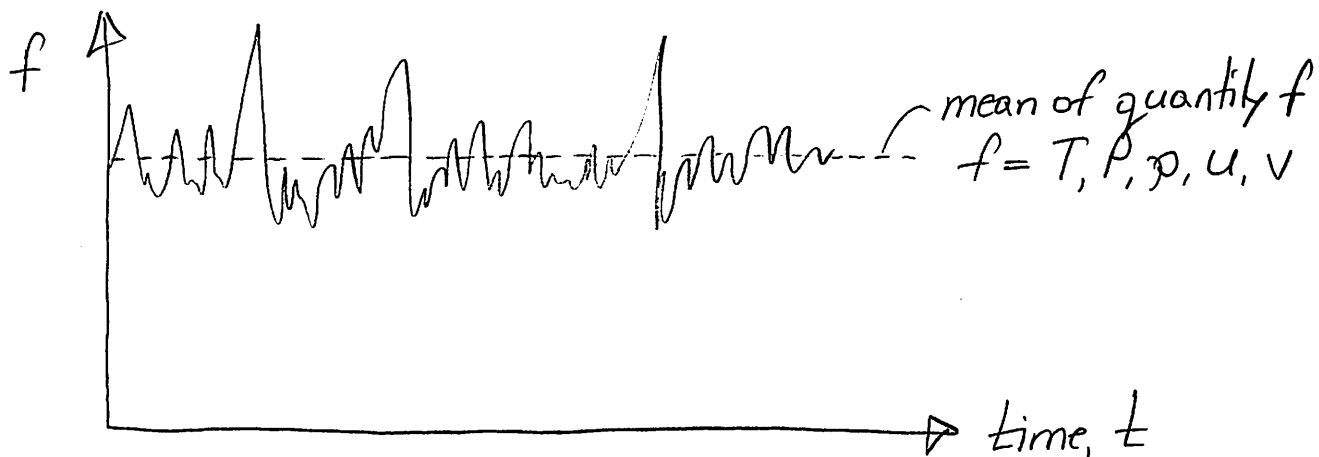
Turbulent Flow

Turbulence in a fluid can be seen as a spectrum of coexisting vortices (eddies) in which kinetic energy from larger ones is dissipated to successively smaller ones until the very smallest of these vortices are damped out by viscous shear.

For a flat plate, external flow:



We handle this chaos by time averaging. Turbulent flow causes fluctuations of the velocity components, pressure, temperature, and in compressible flow, density.

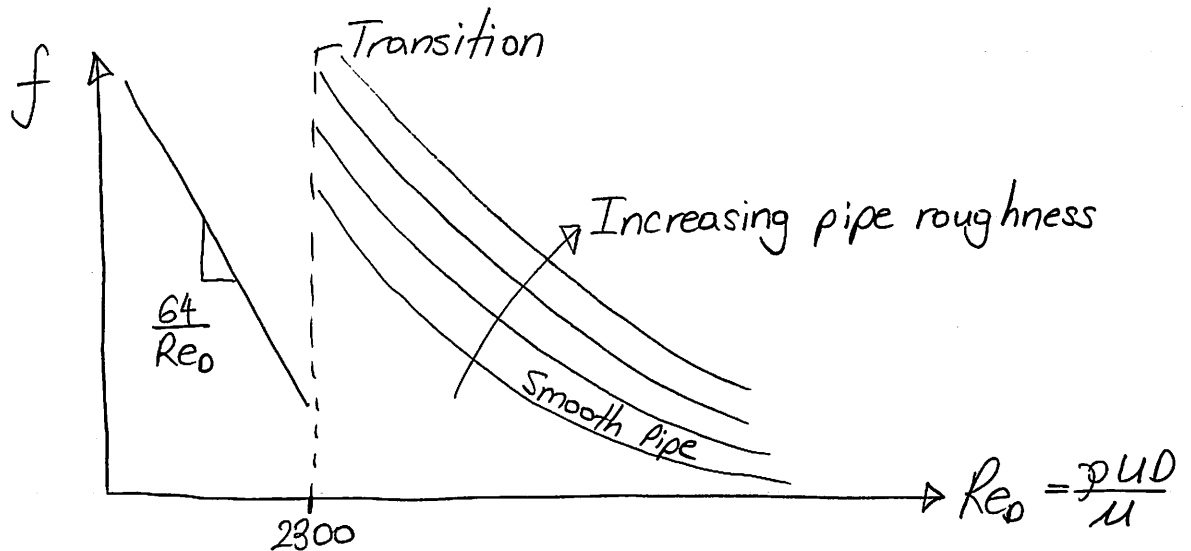


By defining $f = \bar{f} + f'$, we can back substitute into our momentum equations & solve (ME 420) (131)

$\uparrow$ $\uparrow$
 Mean Fluctuating
 Component Component

We handle the mathematics in ME420 & 521, for 320, we learn to use the results: (correlations)

One of the best & useful results is the empirical Moody chart:



Good correlation to know (if you have a smooth pipe)

$$f = (0.79 \ln(Re_D) - 1.64)^{-2} \quad 10^4 < Re_D < 10^6, \text{ smooth pipe}$$

$$f = \frac{\Delta P}{\left(\frac{L}{D}\right) \frac{1}{2} \rho U^2}, \quad C_f = \frac{\bar{\tau}_0}{\frac{1}{2} \rho U^2}, \quad C_f = \frac{f}{4} \quad (\text{for smooth pipes})$$

L = pipe length, D = pipe diameter, U = average flow velocity
 $\bar{\tau}_0$ = average shear stress at the pipe wall

You may ask, why friction & pressure drop, this isn't fluid mechanics. Well, momentum transfer & heat transfer are intricately linked, hence to solve for heat transfer, we need fluid mechanics, as we will see in the Nusselt # correlation

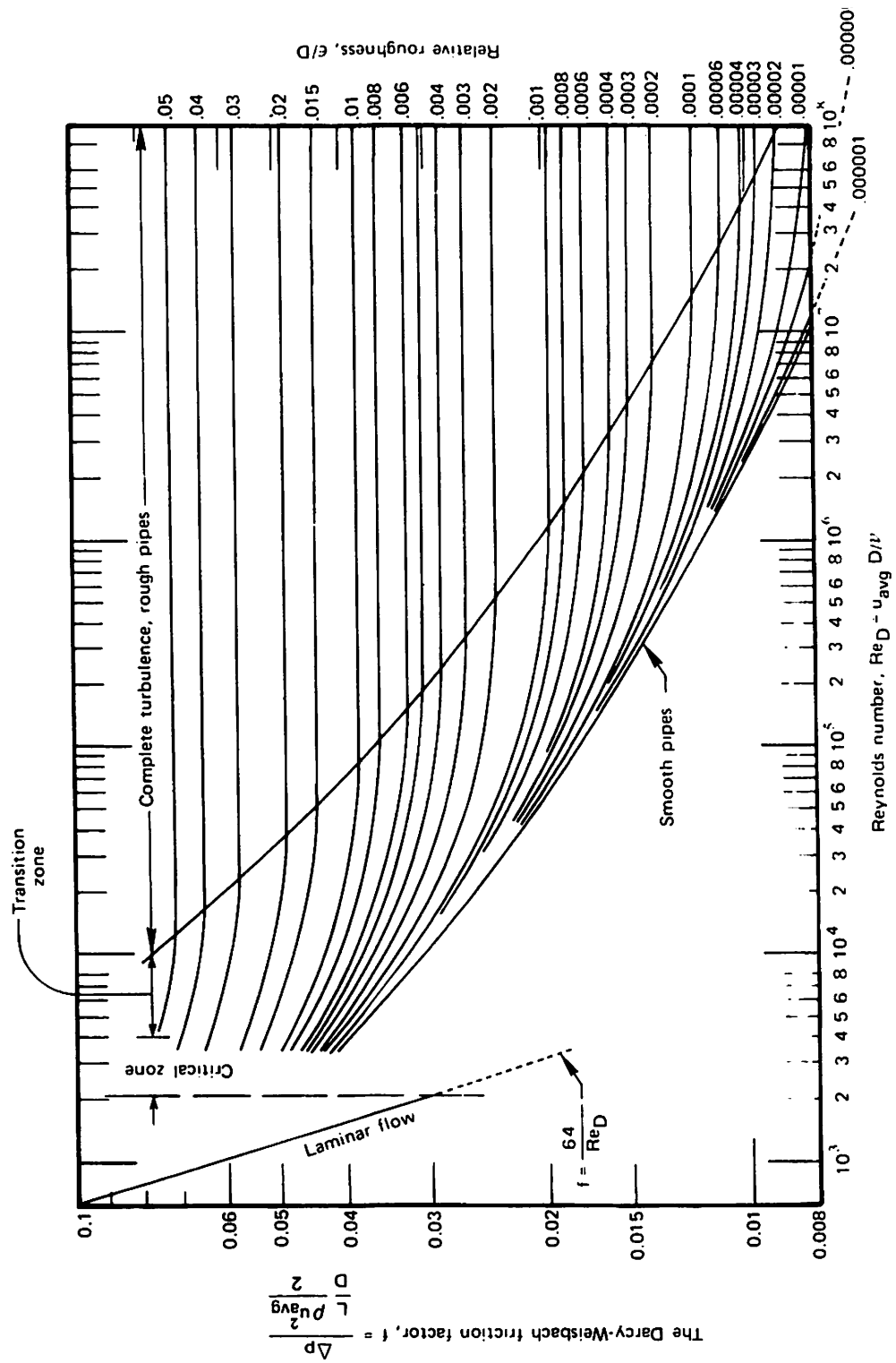


Figure 7.6 Pipe friction factors.

* Adapted from Lienhard & Lienhard, "A Heat Tre

Turbulent Flow Heat Transfer (Internal flow)

Some very good correlations to know are:

$$\boxed{\overline{Nu}_0 = \frac{\bar{h}D}{k_f} = 0.023 Re_0^{0.8} Pr^{0.4}} \Rightarrow \text{Dittus - Boelter eq.} \\ \Rightarrow Re_0 > 10,000$$

For a more accurate result, use:

$$\boxed{Nu_0 = \frac{\bar{h}D}{k_f} = \frac{(f/8)(Re_0 - 1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)}} \Rightarrow \text{Gnielinski correlation} \\ 3000 < Re_0 < 5 \times 10^6$$

External Flow (Turbulent heat transfer)

Remember, $Re_{CR} = \frac{V_\infty X_{CR}}{\nu} = 5 \times 10^5$ (Turbulent transition)

Some useful correlations: ($\tau_{0,x}$ = local shear stress at x)

$$\boxed{C_{f,x} = \frac{\tau_{0,x}}{\frac{1}{2}\rho V_\infty^2} = 0.0592 Re_x^{-1/5}} \quad 10^5 < Re_x < 10^7 \\ \text{Local skin friction coefficient at } x.$$

$$\boxed{C_{f,x} = \frac{\tau_{0,x}}{\frac{1}{2}\rho V_\infty^2} = 0.026 Re_x^{-1/7}} \quad 10^7 < Re_x < 10^9$$

Note, these external flow correlations are also on pg. 123 of the notes (Table 7.7)

For even greater accuracy: $\boxed{C_{f,x} = \frac{0.455}{\ln(0.06 Re_x)^2}} \quad 10^5 < Re_x < 10^9$
 $\hookrightarrow$ White correlation

Typically, we need averaged skin friction coefficient for the whole plate:

$$\tau_{0,x} = \frac{1}{2} \rho V_\infty^2 C_{f,x} = \frac{1}{2} \rho V_\infty^2 \cdot 0.0592 Re_x^{-0.2}$$

$$\overline{\tau_{0,L}} = \frac{1}{L} \int_0^L \tau_{0,x} dx = \frac{\tau_{0,L}}{0.8}$$

$$C_{f,av} = \overline{C_f} = \frac{C_{f,L}}{0.8} = \frac{0.0592 Re_L^{-1/5}}{0.8}$$

$$\boxed{\overline{C_f} = 0.074 Re_L^{-1/5}} \Rightarrow \text{Plate averaged skin friction coefficient.}$$

External Heat Transfer

$$\boxed{Nu_x = \frac{hx}{k_f} = 0.029 Re_x^{0.8} Pr^{1/3}} \Rightarrow \text{Local Nusselt \# Analytical Result}$$

$$\boxed{Nu_x = \frac{hx}{k_f} = 0.029 Re_x^{0.8} Pr^{0.43}} \Rightarrow \text{Local Nusselt \# Experimental Result Whitacker Correlation}$$

As we can see, the analytical & experimental results match pretty well.

$$\boxed{\overline{Nu_L} = \frac{\overline{h}L}{k_f} = 0.036 Re_L^{0.8} Pr^{0.43}} \quad \begin{matrix} 0.7 < Pr < 400 \\ 5 \times 10^5 < Re_L < 3 \times 10^7 \end{matrix}$$

For a more accurate correlation:

$$\boxed{Nu_x = \frac{\left(\frac{C_{f,x}}{2}\right) Re_x Pr}{1 + 12.7 \left(\frac{C_{f,x}}{x}\right)^{1/2} (Pr^{2/3} - 1)}} \Rightarrow \text{White Correlation} \quad \begin{matrix} 0.5 < Pr < 2000 \\ 5 \times 10^5 < Re_x < 10^7 \end{matrix}$$